Predicate Logic

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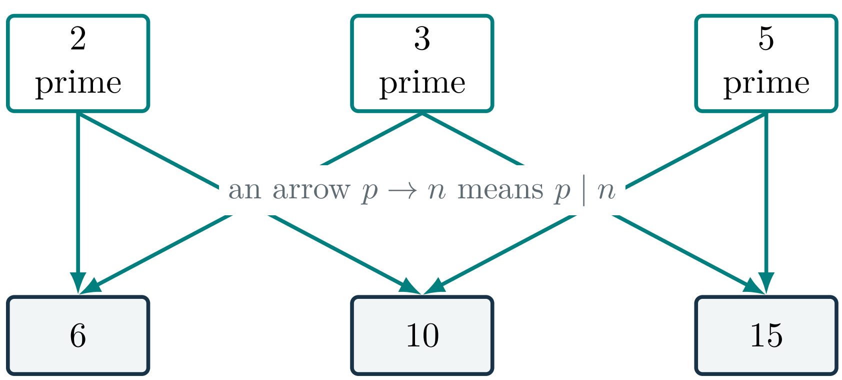
1 Motivation

Propositional logic treats each complete claim as one unit. Let P_n mean "n has a prime divisor." Over $D = \{2, 3, 4\}$, the claim holds everywhere exactly when $P_2 \wedge P_3 \wedge P_4$ is true. Over all integers greater than 1, however, the list $P_2 \wedge P_3 \wedge P_4 \wedge \cdots$ never ends. Propositional logic has no variable that can stand for an arbitrary integer.

Predicate logic adds that reusable structure. Over the domain $\mathbb{Z}$ of all integers, let $H(n)$ mean "n has a prime divisor." The familiar theorem "Every integer greater than 1 has a prime divisor" is written as $\forall n\,(n > 1 \to H(n))$. One finite statement now describes a pattern across the entire domain. This gain in expressive power is why we need predicate logic; the next sections unpack the preview and replace $H(n)$ by its exact definition.

Here *predicate logic* means first-order logic: variables range over objects in a domain, not over predicates themselves.



The picture shows only three sample integers, not a proof of the theorem. Each displayed integer has an incoming arrow from a prime divisor. Yet no one displayed prime divides all three integers. Predicate logic will make both ideas precise: each input may have a suitable prime even when no single prime works for all inputs.

This language appears throughout mathematics and computing: definitions describe which objects have a property, theorems quantify over objects, database queries search for matching objects, and software specifications constrain every permitted execution.

2 Language of predicate logic

2.1 Predicates and domains

We use $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ for the set of all integers and $\mathbb{R}$ for the set of all real numbers.

A **predicate** is a statement template containing variables. For example,

$$\mathsf{Prime}(x), \qquad \mathsf{Even}(x), \qquad d \mid n$$

mean that x is prime, x is even, and d divides n, respectively. The divisibility relation has two argument positions, and their order matters: $2 \mid 6$ is true, while $6 \mid 2$ is false.

The **domain** or **universe of discourse** is the set of objects over which variables range. A formula has no fixed meaning until its domain and predicates are understood.

Over the domain $\mathbb{Z}$, the expression $\mathsf{Prime}(x)$ is an **open formula**: its truth depends on the value of the free variable x. Now add a quantifier:

$$\exists x \in \mathbb{Z}\ \mathsf{Prime}(x).$$

Read this as "There exists an integer x such that x is prime." The domain $\mathbb{Z}$ says what x may denote; the quantifier $\exists$ **binds** occurrences of x in its scope $\mathsf{Prime}(x)$. No variable remains free, so the whole expression is a proposition.

If the domain has already been declared to be $\mathbb{Z}$, we usually shorten the formula to $\exists x\, \mathsf{Prime}(x)$. With a larger background domain, the bounded notation $\exists x \in \mathbb{Z}\ \mathsf{Prime}(x)$ abbreviates $\exists x\,(x \in \mathbb{Z} \wedge \mathsf{Prime}(x))$.

For example, the open formula $x^2 \geq x$ holds for every $x \in \mathbb{Z}$ but not for every $x \in \mathbb{R}$: $x = \frac{1}{2}$ is a real counterexample. The formula did not change; the domain did.

Unless stated otherwise, this course assumes that domains are nonempty.

2.2 Universal and existential quantifiers

The **universal quantifier** states that a property holds throughout the domain:

$$\forall x\, P(x) \qquad \text{"for every } x,\ P(x) \text{ holds."}$$

The **existential quantifier** states that at least one object has the property:

$$\exists x\, P(x) \qquad \text{"there is an } x \text{ for which } P(x) \text{ holds."}$$

Existence means one or more, not exactly one.

If a finite domain is $D = \{a_1, \dots, a_n\}$, then

$$\begin{aligned} \forall x\, P(x) &\quad \text{behaves like} \quad P(a_1) \wedge \cdots \wedge P(a_n), \\ \exists x\, P(x) &\quad \text{behaves like} \quad P(a_1) \vee \cdots \vee P(a_n). \end{aligned}$$

This gives a useful evidence table.

Claim	To establish it	To refute it
$\forall x\, P(x)$	reason about an arbitrary x	give one counterexample
$\exists x\, P(x)$	find one x for which $P(x)$ is true	show every possible x fails

For $D = \{1, 2, 3, 4\}$, the statement $\forall x\,(x < 5)$ is true because every case works. The statement $\forall x\,(x \text{ is prime})$ is false because 4 is a counterexample. The statement $\exists x\,(x \text{ is even})$ is true because 2 makes the condition true.

Before manipulating symbols, ask what the quantifier demands: an arbitrary object, one successful example, a counterexample, or failure for every possible value.

2.3 Restricted quantification

Suppose the domain is the integers. A phrase such as "every prime greater than 2" does not change the domain. It restricts which integers must satisfy the conclusion.

English pattern	Formula over a larger domain
Every prime greater than 2 is odd	$\forall n\,((\mathsf{Prime}(n) \wedge n > 2) \to \mathsf{Odd}(n))$
Some prime divides 12	$\exists p\,(\mathsf{Prime}(p) \wedge p \mid 12)$
No odd integer is divisible by 2	$\forall n\,(\mathsf{Odd}(n) \to 2 \nmid n)$
Some integer greater than 1 is not prime	$\exists n\,(n > 1 \wedge \neg\mathsf{Prime}(n))$

The two central patterns are worth remembering for their meaning, not only their shape:

- **Universal restriction uses implication.** If an integer is a prime greater than 2, then it must be odd. Other integers impose no requirement.
- **Existential restriction uses conjunction.** The chosen number must be both prime and divide 12; 2 and 3 both work.

For example, $\exists p\,(\mathsf{Prime}(p) \wedge p \mid 12)$ reads: "There is an integer p such that p is prime and p divides 12."

The prime-divisor theorem can now be written over the domain of integers:

$$\forall n\Big(n > 1 \to \exists p\big(\mathsf{Prime}(p) \wedge p \mid n\big)\Big).$$

Read aloud from the outside inward: "For every integer n, if $n > 1$, then there exists a prime p that divides n."

2.4 Free and bound variables

A quantifier binds occurrences of its variable within its **scope**. Parentheses make that scope visible.

In

$$n > 1 \wedge \exists p\,(\mathsf{Prime}(p) \wedge p \mid n),$$

p is bound by $\exists p$, while n is free. The expression is still open. In

$$\forall n\Big(n > 1 \to \exists p\,(\mathsf{Prime}(p) \land p \mid n)\Big),$$

both variables are bound, so the formula is a proposition.

The name of a bound variable is local. Renaming it does not change meaning:

$$\exists p\,(\mathsf{Prime}(p) \land p \mid n) \equiv \exists q\,(\mathsf{Prime}(q) \land q \mid n).$$

Renaming is safe only when the new name does not accidentally capture a free variable. A reliable habit is to give overlapping quantifiers different variable names before transforming a formula.

2.5 Quantifier negation

The negation of "every object works" is "some object fails." The negation of "some object works" is "every object fails." Formally,

$$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x),$$
$$\neg\exists x\, P(x) \equiv \forall x\, \neg P(x).$$

These are De Morgan's laws for quantifiers. The finite-domain interpretation explains the pattern: negating a long conjunction produces a disjunction of negations, and vice versa.

Negation example

Negate the prime-divisor theorem and move all negations inside the predicates.

Let $H(n) := \exists p(\mathsf{Prime}(p) \land p \mid n)$. In words, $H(n)$ says that "n has a prime divisor." Thus $\forall n(n > 1 \to H(n))$ says that "every integer greater than 1 has a prime divisor." Starting from the formal statement, justify one transformation per line:

$$\begin{aligned}
\neg\forall n(n > 1 \to H(n)) &\equiv \exists n\neg(n > 1 \to H(n)) && \text{negate } \forall;\\
&\equiv \exists n(n > 1 \land \neg H(n)) && \text{negate the implication;}\\
&\equiv \exists n\Big(n > 1 \land \forall p\neg(\mathsf{Prime}(p) \land p \mid n)\Big) && \text{expand } H \text{ and negate } \exists;\\
&\equiv \exists n\Big(n > 1 \land \forall p(\neg\mathsf{Prime}(p) \lor p \nmid n)\Big) && \text{use } \neg(A \land B) \equiv \neg A \lor \neg B;\\
&\equiv \exists n\Big(n > 1 \land \forall p(\mathsf{Prime}(p) \to p \nmid n)\Big) && \text{rewrite } \neg A \lor B \text{ as } A \to B.
\end{aligned}$$

In English: *some integer greater than* 1 *has no prime divisor*. This negated statement is false over the integers: every integer greater than 1 does have a prime divisor. Negation has converted a universal theorem into a search for one counterexample.

3 Quantifier order

Let $\mathsf{Opens}(k, \ell)$ mean "key k opens lock ℓ." Suppose each lock has a key, but different locks require different keys. Then

$$\forall \ell\, \exists k\, \mathsf{Opens}(k, \ell)$$

is true: after a lock ℓ is chosen, its key k may depend on that lock. It does not imply

$$\exists k\,\forall \ell\ \mathsf{Opens}(k, \ell).$$

The second claim requires one key that opens every lock. This invalid inference mistakes separate local choices for one global choice.

This example gives the operational distinction:

$$\begin{aligned} \forall x \exists y\, R(x, y): &\quad \text{for each } x\text{, a possibly different } y \text{ may be chosen,} \\ \exists y \forall x\, R(x, y): &\quad \text{one } y \text{ must work for every } x. \end{aligned}$$

Quantifiers of the same type can be reordered:

$$\begin{aligned} \forall x \forall y\, P(x, y) &\equiv \forall y \forall x\, P(x, y), \\ \exists x \exists y\, P(x, y) &\equiv \exists y \exists x\, P(x, y). \end{aligned}$$

Mixed quantifiers cannot simply be swapped. There is, however, a valid one-way implication:

$$\exists y \forall x\, R(x, y) \quad \Longrightarrow \quad \forall x \exists y\, R(x, y).$$

If one y works for every x, then for each x we may choose that same y. The converse is generally false:

$$\forall x \exists y\, R(x, y) \quad \not\Rightarrow \quad \exists y \forall x\, R(x, y).$$

Separate choices for individual values of x need not give one global choice. The reliable question is: may the later choice depend on the earlier one?

4 Translation

Use the following procedure when translating English into predicate logic.

1. **State the domain.** Say exactly what each variable can denote.
2. **Define predicates.** Include the argument order of every relation.
3. **Determine the choices.** Which object is selected first, and may later choices depend on it?
4. **Write the formula.** Use implication for universal restrictions and conjunction for existential requirements.
5. **Read back.** Translate the formula into English and test it on a small example.

Existence and uniqueness

Translate "Every even integer has exactly one integer half" over the domain of integers.

For a fixed even integer n, existence says

$$\exists k\,(n = 2k).$$

To express uniqueness, require every other candidate j satisfying $n = 2j$ to equal k. Combining both parts gives

$$\forall n\,(\mathsf{Even}(n) \to \exists k\,(n = 2k \land \forall j\,(n = 2j \to j = k)))\,.$$

The reusable pattern is:

$$\text{exactly one value works} = \text{at least one works} + \text{any value that works is that one.}$$

The symbol $\exists!y$ is often used as an abbreviation for this pattern, but the expanded formula shows what uniqueness means.

More generally, write "exactly one y satisfies $C(y)$" in two steps:

1. **Existence:** choose a candidate y and require $C(y)$.
2. **Uniqueness:** consider every possible candidate z that satisfies the same condition and require $z = y$.

Combining the steps gives

$$\exists y\Big(C(y) \land \forall z\big(C(z) \to z = y\big)\Big).$$

For example, "every integer has exactly one additive inverse" becomes

$$\forall x \in \mathbb{Z}\; \exists y \in \mathbb{Z}\,(x + y = 0 \land \forall z \in \mathbb{Z}\,(x + z = 0 \to z = y))\,.$$

Mathematical writing

Formal logic exposes the structure of a claim, but mathematical papers and textbooks rarely use only words or only symbols. They normally use a middle register. Consider three versions of Goldbach's conjecture, taking all variables over the integers:

Fully verbal: Every even integer greater than two is the sum of two prime numbers.

Fully symbolic: $\forall n\big((\mathsf{Even}(n) \land n > 2) \to \exists p \exists q(\mathsf{Prime}(p) \land \mathsf{Prime}(q) \land n = p + q)\big)$.

Typical mathematical prose: Every even integer $n > 2$ can be written as $n = p + q$ for primes p and q.

The last version is the usual register. Phrases such as "even integer" and "for primes p and q" do work that a fully symbolic formula assigns to predicates and quantifiers. When the context is clear, this mixed style remains precise while being much easier to read. Good mathematical writing uses words for meaning and notation for structure.

5 Equivalence and normal form

5.1 Quantified equivalences

Two quantified formulas are logically equivalent when they agree for every interpretation over the same domain.

Two distribution laws are valid:

$$\forall x\,(P(x) \wedge Q(x)) \equiv (\forall x\,P(x)) \wedge (\forall x\,Q(x)),$$
$$\exists x\,(P(x) \vee Q(x)) \equiv (\exists x\,P(x)) \vee (\exists x\,Q(x)).$$

The first law says that "every object has both properties P and Q" is the same claim as "every object has P, and every object has Q." The second says that "some object has property P or Q" is the same claim as "some object has P, or some object has Q." Reading both sides aloud makes the equivalences visible without expanding the domain element by element.

The crossed patterns are valid in one direction:

$$\exists x\,(P(x) \wedge Q(x)) \Longrightarrow (\exists x\,P(x)) \wedge (\exists x\,Q(x)),$$
$$(\forall x\,P(x)) \vee (\forall x\,Q(x)) \Longrightarrow \forall x\,(P(x) \vee Q(x)).$$

If one object satisfies both properties, then each separate existence claim is true. Similarly, if one property holds for every object, then every object has at least one of the two properties.

The converse directions are not valid in general:

$$(\exists x\,P(x)) \wedge (\exists x\,Q(x)) \not\Rightarrow \exists x\,(P(x) \wedge Q(x)),$$
$$\forall x\,(P(x) \vee Q(x)) \not\Rightarrow (\forall x\,P(x)) \vee (\forall x\,Q(x)).$$

Parity gives a direct example over the integers. Every integer is even or odd, so

$$\forall n\,(\mathsf{Even}(n) \vee \mathsf{Odd}(n))$$

is true. But "every integer is even" is false, and "every integer is odd" is false. Therefore

$$\forall n\,(\mathsf{Even}(n) \vee \mathsf{Odd}(n)) \quad \not\Rightarrow \quad (\forall n\,\mathsf{Even}(n)) \vee (\forall n\,\mathsf{Odd}(n)).$$

Similarly, some integer is even and some integer is odd: 2 and 3 make the two separate claims true. No integer is both even and odd. Hence

$$(\exists n\,\mathsf{Even}(n)) \wedge (\exists n\,\mathsf{Odd}(n)) \quad \not\Rightarrow \quad \exists n\,(\mathsf{Even}(n) \wedge \mathsf{Odd}(n)).$$

To disprove a claimed equivalence, do not attempt a general proof. Build the smallest interpretation in which the two sides have different truth values.

5.2 Prenex normal form

Begin with a concrete example:

$$\underbrace{\forall x\,\exists y}_{\text{quantifier prefix}}\ \underbrace{(P(x) \vee R(x,y))}_{\text{quantifier-free matrix}}.$$

This formula is in **prenex normal form** (PNF): all of its quantifiers appear at the front, and the remaining part contains no quantifiers. PNF gives a standard shape for comparing formulas and for later automated reasoning. It is not a unique spelling of a formula.

When x is not free in A, a quantifier can be pulled across $\wedge$ or $\vee$. Under our nonempty-domain convention, the following pattern is valid for $Q \in \{\forall, \exists\}$ and $\circ \in \{\wedge, \vee\}$:

$$A \circ Qx\, P(x) \equiv Qx\, (A \circ P(x)).$$

The side condition matters. If x is free in A, extending the quantifier's scope changes which occurrences it binds.

Use this conversion procedure:

1. Rename bound variables so different quantifiers use different names.
2. Eliminate $\rightarrow$ and $\leftrightarrow$.
3. Move negations inward, switching $\forall$ and $\exists$ when crossed.
4. Pull quantifiers outward, checking the free-variable side condition at each step.

PNF conversion

Convert

$$\exists z \Big(\exists y\, R(y, z) \rightarrow \forall y\, Q(y, z) \Big)$$

to prenex normal form.

First rename the second bound variable:

$$\begin{aligned}
&\exists z \Big(\exists y\, R(y, z) \rightarrow \forall x\, Q(x, z) \Big) && \\
&\equiv \exists z \Big(\neg \exists y\, R(y, z) \vee \forall x\, Q(x, z) \Big) && \text{remove the implication} \\
&\equiv \exists z \Big(\forall y\, \neg R(y, z) \vee \forall x\, Q(x, z) \Big) && \text{move negation inward} \\
&\equiv \boxed{\exists z \forall y \forall x\, (\neg R(y, z) \vee Q(x, z))} && \text{pull quantifiers outward.}
\end{aligned}$$

The formula after the quantifiers is quantifier-free. Because the last two quantifiers are both universal, their order could be reversed without changing meaning.

6 Problem solving

Checklist

1. Write the domain and predicate definitions before the formula.
2. Mark the scope of every quantifier and locate any free variable.
3. Read nested quantifiers as a sequence of choices.
4. For an existential claim, find one value that works; against a universal claim, find one counterexample.
5. When negating, switch each crossed quantifier and continue until negation reaches an atomic predicate.

6. When testing equivalence, use laws for a proof or a tiny interpretation for a disproof.
7. When converting to PNF, rename variables before moving quantifiers.
8. Read the final formula back in English as a last check.

Practice

Assume nonempty domains throughout.

1. Over the integers, determine the truth of $\forall x \exists y\,(x+y=0)$ and $\exists y \forall x\,(x+y=0)$.
2. Over the integers, translate "Every integer greater than 1 has a prime divisor."
3. Negate $\forall n\,(\mathsf{Even}(n) \to \exists k\,(n=2k))$ with no negation before a quantifier.
4. Over the integers, explain why $(\exists n\,\mathsf{Even}(n)) \wedge (\exists n\,\mathsf{Odd}(n))$ does not imply $\exists n(\mathsf{Even}(n) \wedge \mathsf{Odd}(n))$.
5. Explain which variables are free in $P(x) \wedge \exists y\,R(x,y)$.
6. Expand "There is exactly one x such that $P(x)$" without using $\exists!$.
7. Convert $\neg\exists x\,(P(x) \to \forall y\,Q(x,y))$ to PNF.

Answers

1. The first is true: choose $y=-x$. The second is false: no single integer is the additive inverse of every integer.
2. $\forall n\big(n>1 \to \exists p(\mathsf{Prime}(p) \wedge p \mid n)\big)$.
3. $\exists n\big(\mathsf{Even}(n) \wedge \forall k\,(n \neq 2k)\big)$.
4. The separate existence claims are true because 2 is even and 3 is odd. The conclusion is false because no integer is both even and odd.
5. x is free; y is bound.
6. $\exists x\big(P(x) \wedge \forall y(P(y) \to y=x)\big)$.
7.

$$\begin{aligned}
\neg\exists x(P(x) \to \forall y Q(x,y)) &\equiv \forall x \neg(\neg P(x) \vee \forall y Q(x,y)) \\
&\equiv \forall x(P(x) \wedge \exists y \neg Q(x,y)) \\
&\equiv \forall x \exists y(P(x) \wedge \neg Q(x,y)).
\end{aligned}$$

7 Summary

- Predicates describe properties and relations over a stated domain.
- Quantifiers bind variables and determine whether every value must work or one successful example is enough.

- Quantifier order determines whether one choice may depend on another, while scope controls meaning.
- Negation, counterexamples, and PNF transformations must preserve that structure.

The next lecture studies inference and proof: deriving conclusions from quantified premises.